
ASSIGNMENT

PROGRAMMING USING C

WEEK – 11

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DEPT & SEC :ECE F

Week – 11

Two strings **A** and **B** comprising of lower case English letters are compatible if they are equal or can be made equal by following this step any number of times:

```
#include<stdio.h>
#include<string.h>
int main()
{
    char str1[1000000],str2[1000000];
    int flag=1;
    scanf("%s",str1);
    scanf("%s",str2);
    int a=strlen(str1);
    int b=strlen(str2);
    if(a==b)
    {
        for(int i=a-1;i<=0;i--)
        {
            while(str1[i]!=str2[i])
            {
                for(int j=0;j<=i;j++)
                {
                    if(str1[j]<'z')
                        str1[j]++;
                    else
                    {
                        flag=0;
                        break;
                    }
                    if(flag==0)
                        break;
                }
            }
        }
    }
    else
        flag=0;

    if(flag==0)
        printf("NO");
    else
        printf("YES");
    return 0;
}
```

	Input	Expected	Got	
✓	abaca cdbda	YES	YES	✓

Passed all tests! ✓

Danny has a possible list of passwords of Manny's facebook account. All passwords length is odd. But Danny knows that Manny is a big fan of palindromes. So, his password and reverse of his password both should be in the list.

```
#include<stdio.h>
#include<string.h>
int main()
{
    int n,flag=0;
    char temp;
    scanf("%d",&n);
    char words[n][14];
    for(int i=0;i<=n;i++)
    scanf("%s",words[i]);
    char reverse[14];
    for(int i=0;i<n-1;i++)
    {
        strcpy(reverse,words[i]);
        int size=strlen(reverse);

        for(int k=0;k<size/2;k++)
        {
            temp=reverse[k];
            reverse[k]=reverse[size-k-1];
            reverse[size-k-1]=temp;
        }
        for(int j=i+1;j<n;j++)
        {
            if(strcmp(reverse,words[j])==0)
            {
                flag=1;
                break;
            }
        }
        if(flag==1)
            break;
    }
    int len=strlen(reverse);
    printf("%d %c",len,reverse[len/2]);
    return 0;
}
```

	Input	Expected	Got	
✓	4 abc def feg cba	3 b	3 b	✓

Passed all tests! ✓

Joey loves to eat Pizza. But he is worried as the quality of pizza made by most of the restaurants is deteriorating. The last few pizzas ordered by him did not taste good :(Joey is feeling extremely hungry and wants to eat pizza. But he is confused about the restaurant from where he should order. As always he asks Chandler for help.

```
#include<stdio.h>
#include<string.h>
int main()
{
    int n;
    scanf("%d",&n);
    char res[n][21];
    int rate[n];
    for(int i=0;i<n;i++)
    {
        scanf("%s",res[i]);
        scanf("%d",&rate[i]);
    }
    int max=rate[0];
    char ans[20];
    strcpy(ans,res[0]);
    for(int i=1;i<n;i++)
    {
        if(rate[i]>max)
        {
            max=rate[i];
            strcpy(ans,res[i]);
        }
        else if(rate[i]==max)
        {
            if(strcmp(res[i],ans)<0)
                strcpy(ans,res[i]);
        }
    }
    printf("%s",ans);
    return 0;
}
```

	Input	Expected	Got	
✓	3 Pizzeria 108 Dominos 145 Pizzapizza 49	Dominos	Dominos	✓

Passed all tests! ✓

These days Bechan Chacha is depressed because his crush gave him list of mobile number some of them are valid and some of them are invalid. Bechan Chacha has special power that he can pick his crush number only if he has valid set of mobile numbers. Help him to determine the valid numbers.

```
#include<stdio.h>
#include<string.h>
int main()
{
    int t;
    scanf("%d",&t);
    while(t-->0)
    {
        int flag=1;
        char s[100000];
        scanf("%s",s);
        int k=strlen(s);

        if(k==10)
        {
            for(int i=0;i<10;i++)
            {
                if(s[i]!='0')
                {
                    flag=0;
                    break;
                }
                if(s[i]<'0' || s[i]>'9')
                {
                    flag=0;
                    break;
                }
            }
        }
        else
        {
            flag=0;
        }
        if(flag==1)
            printf("YES\n");
        else
            printf("NO\n");
    }
    return 0;
}
```

	Input	Expected	Got	
✓	3	YES	YES	✓
	1234567890	NO	NO	
	0123456789	NO	NO	
	0123456.87			

Passed all tests! ✓